Surface Water Quality Index Estimation via Gradient Tree Boosting and Stoichiometric Feature Engineering

Semester 5 Applied Machine Learning (AML) | Project 18: Water Quality Prediction

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Repository: **https://github.com/MAYANKSHARMA01010/Semester_5_AML_DL**

***Abstract*—Surface water quality monitoring is essential for public health, biodiversity, and municipal water security. Traditional laboratory assays, such as 5-day Biochemical Oxygen Demand (BOD5), suffer from a 120-hour incubation latency. This paper presents Phase 1 of an applied machine learning investigation predicting the Canadian Council of Ministers of the Environment Water Quality Index (CCME WQI) across 2,827,977 multi-national sensor observations spanning 83 years (1940–2023) across 5 nations (England, USA, Ireland, China, and Canada). A data audit disproved legacy claims of missingness, confirming 100% complete-case ground truth. Thermodynamic boundary screening purged 8,360 corrupted telemetry entries (0.296%), yielding 2,819,617 clean observations. We engineered 38 domain features incorporating non-linear log(1+x) transformations, aquatic stoichiometry (Redfield N:P ratio, Benson-Krause DO saturation, Emerson un-ionized ammonia), and chronological station lags. Benchmarked across a stratified 70/15/15 partition, Random Forest (R^2 = 0.9999, MAE = 0.052) and XGBoost (R^2 = 0.9996, MAE = 0.137) dramatically outperformed the mean baseline (MAE = 14.130). Game-theoretic TreeSHAP feature attribution identified orthophosphate as the dominant predictive driver (accounting for >88% of split decisions). We formalize a production modular package (src/) validated by a 12-test automated unit test suite (pytest) and provide an equitable six-member contribution matrix meeting Phase 1 CSAI evaluation requirements.**

I. INTRODUCTION

Freshwater security is under severe anthropogenic pressure from agricultural nutrient runoff, sewage discharge, and industrial contamination. Standard monitoring workflows rely on discrete physical grab sampling and wet-lab analytical assays. However, assays such as BOD5 require multi-day microbiological incubation, preventing immediate detection of acute chemical spills.

Machine learning offers high-throughput surrogate inference, translating real-time probe telemetry directly into continuous indices. While tree ensembles are effective on tabular data, naive models suffer from: (i) sensor corruption artifacts, (ii) heavy nutrient skewness spanning four orders of magnitude, (iii) circular target leakage, and (iv) uninterpretable decision boundaries. Our Phase 1 pipeline explicitly overcomes each of these challenges.

II. RELATED WORK & GAP ANALYSIS

Classical models relied on differential equations like the Streeter-Phelps oxygen sag formula (1925), which require expensive bathymetric tuning and fail to scale across basins. Breiman (2001) introduced Random Forests, demonstrating variance reduction via bootstrap bagging. Chen & Guestrin (2016) developed XGBoost, using regularized second-order Taylor gradients for rapid tabular modeling. Lundberg & Lee (2017) formalized TreeSHAP for exact polynomial-time cooperative game-theoretic feature attribution. We resolve key literature gaps by enforcing thermodynamic screening, preventing target leakage, and capturing non-lookahead station memory.

III. MATHEMATICAL FORMULATION

A. CCME Water Quality Index Formulation

The CCME WQI synthesizes multi-parameter compliance into a continuous score bounded in [0, 100]:

$$\text{CCME WQI} = 100 - \left[\sqrt{(F_1^2 + F_2^2 + F_3^2)} / 1.732 \right]$$

where $1.732 = \sqrt{3}$ normalizes the maximum distance from ideal (0,0,0). F_1 (Scope) is the percentage of failed parameters; F_2 (Frequency) is the percentage of failed tests; and F_3 (Amplitude) scales excursion magnitude through the normalized sum of excursions (nse):

$$\text{nse} = (\sum \text{excursion}_i) / \text{N_total}, \quad F_3 = \text{nse} / (0.01 \cdot \text{nse} + 0.01)$$

B. Random Forest & XGBoost Ensembles

Random Forest aggregates B de-correlated trees trained on bootstrap samples: $\hat{y} = (1/B) \sum T_b(x)$. XGBoost minimizes a regularized objective via second-order Taylor approximation:

$$\tilde{L} \approx \sum \left[g_i f_t(x_i) + 0.5\, h_i f_t^2(x_i) \right] + \gamma T_L + 0.5\, \lambda \sum w_j^2$$

yielding optimal leaf weights $w_j^* = -(\sum g_i) / (\sum h_i + \lambda)$.

IV. DATASET AUDIT & PHYSICS SCREENING

We ingested 2,827,977 records from 5 nations. Data auditing verified 0 null entries. We enforced thermodynamic boundaries based on aquatic chemistry:

TABLE I: PHYSICAL SCREENING BOUNDARIES

Parameter	Valid Envelope	Physics Rationale
pH	[3.0, 12.0]	Natural aquatic acid/base limit
Temperature	[0.0°C, 45.0°C]	Surface liquid freshwater state
Dissolved Oxygen	[0.0, 25.0 mg/L]	Henry's law gas solubility limit
Nutrients (BOD, N, P)	≥ 0.0 mg/L	Non-negative concentration

Screening purged exactly 8,360 corrupted records (0.296%), yielding 2,819,617 clean observations.

TABLE II: MULTI-NATIONAL COMPOSITION

Country	Raw Rows	Clean Rows	Share
England	2,367,978	2,361,042	83.74%
USA	240,654	239,890	8.51%
Ireland	201,345	200,980	7.13%
China	12,000	11,985	0.43%
Canada	6,000	5,720	0.20%
Total	**2,827,977**	**2,819,617**	**100.00%**

V. DOMAIN FEATURE ENGINEERING

We constructed 38 features across 7 categories:

1. Benson-Krause DO Saturation: DO_sat(T) is computed via an empirical polynomial of temperature. We derive DO_%sat = (DO/DO_sat)*100 and DO_deficit = max(0, DO_sat − DO).

2. Redfield Stoichiometric Ratio: N:P = Nitrogen / (Orthophosphate + 1e-5), identifying whether phosphorus or nitrogen limits algal growth.

3. Emerson Toxic Ammonia: Quantifies lethal un-ionized NH3 fraction as a function of temperature and pH.

4. log(1+x) Normalization: Applied y = ln(1+x) to skewed nutrients. Linear correlation between orthophosphate and WQI surged from r = −0.66 to **r = −0.85**.

5. Station Memory: Chronological shift-1 station lags (hist_ortho_lag1) achieved rank correlation ρ = −0.78 with WQI with zero forward lookahead leakage.

VI. RESULTS & MODEL BENCHMARKING

Models were benchmarked on 422,943 validation records across continuous metrics (MAE, RMSE, R^2):

TABLE III: VALIDATION BENCHMARK RESULTS

Model	Val MAE	Val RMSE	Val R^2	Latency
Mean Baseline	14.130	17.656	0.0000	0.02 s
XGBoost	0.137	0.374	0.9996	**0.42 s**
Random Forest	**0.052**	**0.208**	**0.9999**	3.85 s

Both models eliminated >99.6% of baseline variance. Random Forest attained the lowest error, while XGBoost offered 9x faster inference (0.42 s for 422k observations ≈ 1 µs per sensor query).

VII. INTERPRETABILITY VIA TREESHAP

Game-theoretic TreeSHAP feature attribution confirmed that **orthophosphate drives over 88% of all decision tree splits**. Because phosphorus is freshwater's critical limiting nutrient, elevated concentrations trigger immediate WQI collapse. Shift-1 station lag (hist_ortho_lag1) ranked second, proving the power of local environmental memory.

VIII. CODE QUALITY & AUTOMATED TESTING

We engineered a modular package (src/) with features, models, metrics, and utils. A 12-test automated unit test suite executed under pytest passes in 1.65 seconds (100% pass rate), verifying thermodynamic boundaries, stoichiometry, anti-leakage quarantine, and model serialization.

IX. TEAM CONTRIBUTION MATRIX (6 MEMBERS)

TABLE IV: SIX-MEMBER ALLOCATION MATRIX

Member	Core Responsibility	Viva Defense Zone
Mayank Sharma (Lead)	Pipeline Architecture, RF & XGBoost, GitHub & Sharding	Pipeline orchestration, tree algorithms, anti-leakage quarantine.
Team Member 2	Literature Review, CCME WQI Math, Regulations	CCME derivation (F1, F2, F3, nse), EPA/WHO standards.
Team Member 3	Multi-National Audit, Bounds, EDA	Data audit (5 nations), physics screening, outlier diagnostics.
Team Member 4	Feature Engineering, Stoichiometry, Skewness	Benson-Krause DO sat, Redfield N:P, Emerson NH3, log(1+x).
Team Member 5	Optimization, Splitting, Benchmarks	Tree hyperparameters, loss functions, MAE/RMSE/R^2 metrics.
Team Member 6	Explainability (TreeSHAP), Calibration, Testing	Shapley axioms, TreeSHAP algorithm, 12-test pytest suite.

X. CONCLUSION

Phase 1 demonstrated that applied machine learning ensembles achieve near-perfect prediction ($R^2 > 0.999$) of surface water quality indices across 2.82 million international records when paired with domain-informed geochemical feature engineering.

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